Unit 4 Quiz: Scatterplots KEY

Name _____

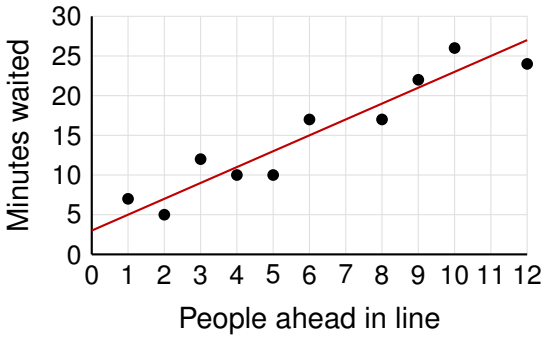
Show your work.

4.1 - Scatterplots and Lines of Best Fit

1. **I describe the direction, form, and strength of a scatterplot, and I choose a good line of best fit.**

   A store counted the people ahead of 10 customers in line and timed each wait.

People ahead in line (x)	1	2	3	4	5	6	8	9	10	12
Minutes waited (y)	7	5	12	10	10	17	17	22	26	24



(a) Describe the direction, form, and strength.

Positive, linear, strong

(b) Draw a line of best fit on the graph.

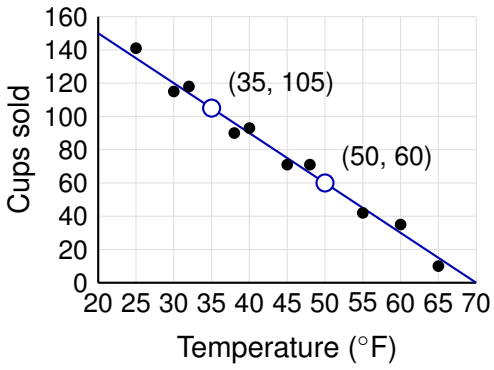
One good line is in red. Accept any line through the middle of the dots.

(c) How many dots are above your line? How many are below?

Red line: 5 above and 5 below

2. **I write the equation of a line of best fit, and I make predictions and know when not to trust them.**

   A school sells hot chocolate at football games. The scatterplot shows the temperature and the number of cups sold at 10 games. A line of best fit is drawn.



(a) Use the two marked points to find the slope.

$$m = \frac{60 - 105}{50 - 35} = \frac{-45}{15} = -3$$

(b) Write the equation of the line.

$105 = -3(35) + b$
$105 = -105 + b$, so $b = 210$
$\hat{y} = -3x + 210$

(c) Predict the number of cups sold when the temperature is 42°F.

$\hat{y} = -3(42) + 210$
$= -126 + 210 = 84$ cups

(d) Predict the number of cups sold when the temperature is 80°F. Does your answer make sense? Explain.

$\hat{y} = -3(80) + 210 = -30$. No. A school cannot sell fewer than 0 cups. 80°F is far outside the data (25°F to 65°F).

4.2 - Slope, Intercept, and Correlation

3. **I explain the slope of a line of best fit in context, and I explain the y-intercept in context, or tell why it has no meaning.**

   A coach recorded how many hours each player practiced last week and how many free throws each player made out of 50. The line of best fit is $\hat{y} = 2x + 15$.

   (a) Explain the slope in context.

   For each additional hour of practice, the predicted number of free throws made increases by 2.

   (b) Explain the y-intercept in context.

   When $x = 0$ (no practice), the predicted number of free throws made is 15.

   A bookstore uses the line $\hat{y} = 0.03x + 4$, where x is the number of pages in a book and y is the price in dollars.

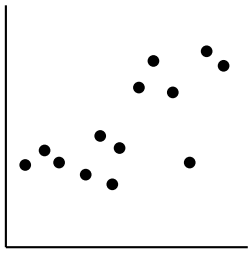
   (c) Does the y-intercept have a real meaning here? Explain.

   No. It says a book with 0 pages costs $4. No book has 0 pages.

4. **I estimate the correlation coefficient r from a scatterplot, and I explain why correlation does not prove causation.**

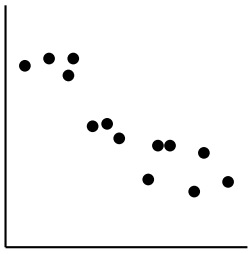
   (a) Match each r to a scatterplot: $r = -0.9$, $r = 0.2$, $r = 0.7$

   A

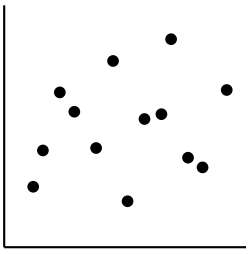
   $r =$ 0.7

   B

   $r =$ −0.9

   C

   $r =$ 0.2

   (b) In many cities, the number of fast food restaurants and the number of schools have a strong positive correlation. Which statement is true? Circle the letter.

   A Building more fast food restaurants causes a city to have more schools.
   B Building more schools causes a city to have more fast food restaurants.
   Ⓒ A lurking variable, such as the number of people in the city, could cause both.
   D There is no relationship, because restaurants and schools are different things.

4.3 - Residuals

5. **I find and explain a residual, and I tell if a point is above or below the line from its residual.**

A taxi company predicts the fare with $\hat{y} = 2.5x + 4$, where x is the distance in miles and y is the fare in dollars.

(a) Fill in the table.

x (miles)	y (actual fare)	$\hat{y}$ (predicted)	residual $y - \hat{y}$
2	10	$2.5(2) + 4 = 9$	$10 - 9 = 1$
4	13	$2.5(4) + 4 = 14$	$13 - 14 = -1$
6	20	$2.5(6) + 4 = 19$	$20 - 19 = 1$
8	22	$2.5(8) + 4 = 24$	$22 - 24 = -2$

(b) Which rides have points above the line? How do you know?

The 2-mile and 6-mile rides. Their residuals are positive.

(c) Explain the residual for the 8-mile ride in context.

The residual is -2. The fare was \$2 less than predicted.

6. **I explain what the least-squares line is, and I use a residual plot to decide if a line is a good model.**

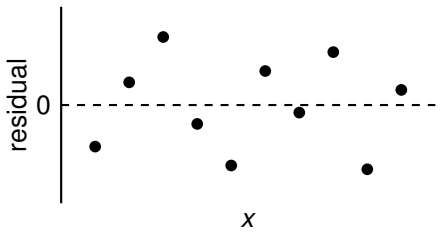
Two lines were fit to the same 4 points. Here are their residuals.

	Residuals				$\Sigma(y - \hat{y})^2$
Line A	2	-3	1	0	$4 + 9 + 1 + 0 = 14$
Line B	-1	2	-2	1	$1 + 4 + 4 + 1 = 10$

(a) Find $\Sigma(y - \hat{y})^2$ for each line. Write it in the table.

(b) Which line is closer to the least-squares line? Explain.

Line B. Its sum of squared residuals (10) is smaller than Line A's (14).



(c) This is the residual plot for a line of best fit for another data set. Which statement is true? Circle the letter.

(A) A line is a good model, because the residuals show no pattern.

B A line is not a good model, because some residuals are negative.

C A line is not a good model, because the residuals are not all 0.

D A line is a good model, because the residuals make a curved pattern.